TRANSPORT TECHNOLOGIES

Digital transmission networks were developed to carry voice and data traffic efficiently over long distances. Before SDH, telephone companies used a technology called the Plesiochronous Digital Hierarchy (PDH) to multiplex many voice channels onto high-capacity transport links. SDH was later introduced to overcome the limitations of PDH by adopting strict synchronization and a more flexible, manageable architecture.

Understanding both PDH and SDH is fundamental to studying modern transport networks, because SDH was designed specifically as the successor to PDH and addresses its shortcomings directly.

Plesiochronous Digital Hierarchy (PDH)

Plesiochronous Digital Hierarchy is a telecommunications technology used to transport large volumes of data over digital transmission equipment such as fibre optic cables and microwave radio systems.

The word *plesiochronous* comes from the Greek words: *plēsios, which is* "near", and *chronos, which* is "time."

This name reflects the key characteristic of PDH: different parts of the network operate with clocks that are nearly, but not perfectly, synchronized.

PDH was standardized in the early 1970s (around 1972) and was the dominant transport technology in telephone networks for several decades.

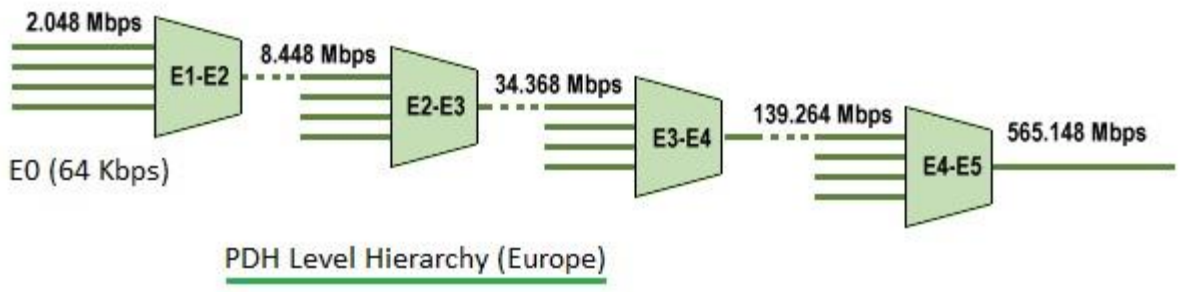
How PDH works

PDH transmits data by time-division multiplexing (TDM), combining multiple lower-speed digital channels into a single higher-speed channel.

The basic channel in PDH is a 64 Kbps digital voice channel (derived from PCM (Pulse Code Modulation), where an analogue voice signal is digitized at 64 Kbps). Multiple 64 Kbps channels are then combined at each level of the hierarchy.

Multiplexing process:

1. 30 individual 64 Kbps voice channels are combined to form a 2 Mbps E1 channel (European standard).
2. Multiple 2 Mbps E1 channels are combined to form an 8 Mbps signal.
3. Multiple 8 Mbps signals are combined to form a 34 Mbps signal.
4. Multiple 34 Mbps signals are combined to form a 140 Mbps signal.



PDH Level Hierarchy (Europe)

In North America, the hierarchy starts at 1.5 Mbps (T1) instead of 2 Mbps (E1), creating a different set of multiplexed rates (T1 → T2 → T3 → T4).

Bit stuffing:

Because the clocks of different channels are not perfectly synchronized, PDH uses a technique called bit stuffing, in which extra bits are inserted into the data stream to align the channels and compensate for timing differences. This bit stuffing is one of the major sources of PDH's complexity.

PDH Hierarchy Table

Level	**European Rate**	**North American Rate**
Base channel	64 Kbps (1 voice channel)	64 Kbps
Level 1	2 Mbps (E1 = 30 channels)	1.5 Mbps (T1 = 24 channels)
Level 2	8 Mbps	6 Mbps (T2)
Level 3	34 Mbps	45 Mbps (T3)
Level 4	140 Mbps	274 Mbps (T4)

Key features of PDH

- Uses point-to-point connections over dedicated lines.
- Networks operate with slightly different clock rates at each node (plesiochronous = near-synchronous).
- Multiplexing is done in a hierarchical, asynchronous manner.
- Well suited for long-distance voice and data links, especially city-to-city or branch-to-branch connections.

Limitations of PDH

PDH has several significant drawbacks that eventually led to the development of SDH:

1. **No direct access to low-rate channels**: To extract a single 2 Mbps channel from a 140 Mbps stream, the entire signal must be fully demultiplexed down through every intermediate level. This is expensive, complex, and wasteful.

2. **Limited network management**: PDH was designed before sophisticated network management was needed; it has very limited OAM (Operations, Administration, and Maintenance) capabilities.
3. **No global standard**: European and North American PDH hierarchies are incompatible, creating problems in international telecommunications.
4. **Lack of synchronization**: Timing differences require bit stuffing, which makes the multiplexing/demultiplexing process more complex and harder to maintain.
5. **No support for future high-speed services**: PDH could not efficiently support emerging broadband services or flexible bandwidth allocation.

Synchronous Digital Hierarchy (SDH)

SDH (Synchronous Digital Hierarchy) is an international standard (defined by the ITU-T) for transmitting large amounts of data over optical fibre and radio links using synchronous multiplexing where all clocks in the network are tightly synchronized to a common reference.

SDH was developed in the late 1980s to replace PDH by addressing all of its major limitations. In North America, the equivalent standard is called **SONET (Synchronous Optical Networking)**. The fundamental principle of SDH is:

Data from multiple tributary sources is byte-interleaved in a synchronous frame structure; because all channels are in fixed, known locations within the frame, individual channels can be added or dropped without fully demultiplexing the entire signal.

How SDH works

SDH uses a hierarchy of continuously repeating, fixed-length frames. Each frame repeats every 125 microseconds (matching the 8000 Hz sampling rate of voice).

The base unit of SDH is the STM-1 (Synchronous Transport Module Level 1) at 155.52 Mbps. Higher levels are exact multiples of STM-1.

SDH rate hierarchy:

SDH Level	Bit Rate	SONET Equivalent
STM-1	155.52 Mbps	OC-3
STM-4	622.08 Mbps	OC-12
STM-16	2,488.32 Mbps (~2.5 Gbps)	OC-48
STM-64	9,953.28 Mbps (~10 Gbps)	OC-192

Because each higher rate is an exact multiple of STM-1, multiplexing and demultiplexing are straightforward and precise no bit stuffing is needed for clock alignment in the SDH core.

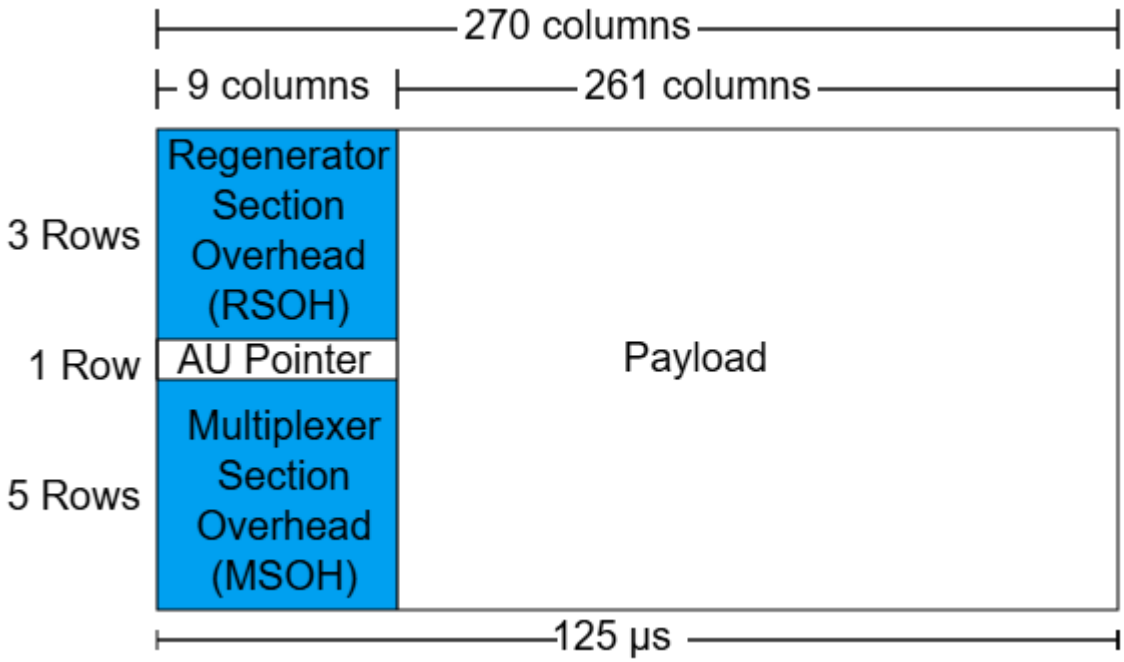
STM-1 Frame Structure

The STM-1 frame is organized as a matrix of 9 rows × 270 columns of bytes, giving a total of 2,430 bytes per frame.

The frame consists of two main parts:

1. **Section Overhead (SOH)**: Contains management, maintenance, and control bytes (for OAM functions).
2. **Synchronous Payload Envelope (SPE)**: The actual payload area where tributary data (PDH signals, ATM cells, IP packets, etc.) is carried.

Within the SPE, individual tributaries can be located directly using pointers, so a single tributary can be extracted without demultiplexing the full STM-1.



SDH multiplexing concept

The SDH multiplexing structure uses several building blocks:

- **VC (Virtual Container)**: The basic unit of payload. PDH tributaries and other signals are mapped into VCs (e.g., VC-12 for a 2 Mbps E1, VC-4 for a 140 Mbps E4).
- **TU (Tributary Unit)**: A VC plus a pointer that indicates the VC's location in the higher frame.
- **TUG (Tributary Unit Group)**: A collection of TUs.

- **AU (Administrative Unit)**: A TUG inside the STM-N frame, with pointer for synchronization.

These structures allow individual tributaries to be added and dropped at any node without full demultiplexing a huge advantage over PDH.

SDH network elements

Network Element	Role
Regenerator	Amplifies and reshapes the optical signal over long distances
Add-Drop Multiplexer (ADM)	Adds or drops individual tributaries from an SDH stream without full demultiplexing
Digital Cross-Connect (DXC)	Switches tributaries between multiple SDH streams, used for network routing and grooming
SDH Terminal Multiplexer	Aggregates PDH tributaries and other signals into an STM-N for transmission

OAM in SDH

One of SDH's key advantages over PDH is its rich **OAM (Operations, Administration, and Maintenance)** capability. The Section Overhead (SOH) bytes carry:

- **Framing bytes**: Identify the start of each STM-1 frame.
- **Performance monitoring bytes**: Track errors and quality in real time (BIP-8 parity checks).
- **Automatic Protection Switching (APS) bytes**: Coordinate fast switchover to backup paths in case of failure (typically within 50 milliseconds).
- **Data Communication Channels (DCC)**: A management network built into the SDH overhead for remote configuration, alarms, and provisioning.

SDH ring protection

SDH networks are often built as rings (like SDH rings or SONET rings):

- **MSP (Multiplex Section Protection)**: Switches traffic to a protect path if the working path fails.
- **SNCP/UPSR (Subnetwork Connection Protection)**: Sends traffic simultaneously on both paths; receiver selects the better signal.
- Protection switching takes place in **< 50 ms**, ensuring high availability for carrier-grade services.

Comparison: PDH and SDH

Feature	PDH	SDH
Synchronization	Plesiochronous (nearly synchronous)	Fully synchronous (global clock)

Multiplexing	Asynchronous, uses bit stuffing	Synchronous, byte-interleaved
Access to tributaries	Full demultiplexing required	Direct add/drop using pointers
OAM capabilities	Very limited	Rich overhead; full OAM
Standards	Different (European E1 vs N.American T1)	Global ITU-T standard (STM-1, STM-N)
Fault recovery	Slow, manual	Fast automatic protection (< 50 ms)
Bit rates	Up to 140 Mbps (European), 274 Mbps (N.Am.)	Up to 9.95 Gbps (STM-64) and beyond
Flexibility	Low; fixed point-to-point	High; ADMs, DXCs, ring topologies
Overhead	Minimal	~4% per STM-1 frame

Relationship between PDH and SDH

SDH was specifically designed to carry PDH traffic over a modern backbone while eliminating PDH's weaknesses. PDH signals (E1, E3, E4, T1, T3) are mapped into Virtual Containers (VCs) inside an SDH frame, so:

- A service provider can still sell E1 or T1 circuits to customers.
- Internally, the network carries and manages those circuits over SDH infrastructure.
- The SDH Add-Drop Multiplexers extract individual E1s directly without touching other traffic.

This backward compatibility was critical for the industry to migrate from PDH to SDH gradually without replacing customer-facing services overnight.

Applications of PDH and SDH

PDH applications:

- Legacy telephone exchange interconnection (still in use in some parts of the world).
- E1/T1 leased line services for business customers.
- Short-haul city or campus interconnection.

SDH applications:

- Long-distance and international backbone networks.
- Mobile network backhaul (GSM, 3G base station connections).
- Carrier-grade VPN and leased-line services.
- As a transport layer under ATM, IP/MPLS, and WDM networks.

FIBER DISTRIBUTED DATA INTERFACE (FDDI)

Fiber Distributed Data Interface (FDDI) is a set of ANSI/ISO standards for high-speed data transmission in LANs and MANs using optical fiber at 100 Mb/s. It was designed mainly as a campus/backbone technology interconnecting multiple lower-speed LANs (Ethernet, Token Ring) over distances up to around 100–200 km.

Main objectives:

- Provide high-bandwidth backbone for large enterprise/campus networks.
- Offer deterministic, bounded-delay access suitable for voice/multimedia and real-time traffic.
- Achieve high reliability and fault tolerance using dual fiber rings.

FDDI standards and OSI position

FDDI is not a single standard but a **family of four specifications** that sit mainly at OSI Layer 1 and the MAC part of Layer 2.

- **FDDI-MAC (Media Access Control):**
  Controls token passing, address recognition, error detection (CRC), token rotation measurement (TRT), and token holding time (THT) calculation.
- **FDDI-PHY (Physical Layer Protocol):**
  Defines encoding, decoding, clocking, and mapping of MAC symbols onto the physical medium.
- **FDDI-PMD (Physical Medium Dependent):**
  Specifies fiber types (multi-mode, single-mode), wavelengths, optical power levels, connectors, distances.
- **FDDI-SMT (Station Management):**
  Handles configuration, ring initialization, fault detection, insertion/removal of stations, and coordination of dual rings.

FDDI sits below the IEEE 802.2 LLC sublayer, which provides a common interface to upper-layer protocols like IP, IPX, etc.

Physical topology and components

Dual counter-rotating rings

FDDI uses two fiber rings: a primary ring for normal data transmission and a secondary ring used either as backup or for extra bandwidth. Traffic on the two rings flows in opposite directions (counter-rotating) to support fault tolerance.

Key physical characteristics:

- Data rate: 100 Mb/s on each ring.

- Maximum number of stations: about 500 on a ring.
- Max ring length: ≈100 km over multimode fiber; with single-mode fiber, individual links can reach tens of kilometers.

Station types and concentrators

Stations are categorized according to how they attach to the rings.

- **Class A / Dual-Attachment Station (DAS):**
  - Connects to both primary and secondary rings (four fibers: two in, two out).
  - Typically routers, backbone servers, or wiring concentrators.
  - Participate in fault-recovery ring wrapping.
- **Class B / Single-Attachment Station (SAS):**
  - Connects through a **wiring concentrator** to only one ring (logically the primary).
  - Used for end-user hosts and departmental servers to simplify cabling and improve robustness.
- **Wiring concentrators:** Provide star-like physical connection but logically maintain a ring; allow easy addition/removal of SAS nodes and help isolate faults.media.

MAC operation: timed token protocol

FDDI's MAC is token-based, similar to IEEE 802.5 Token Ring, but with a **timed token** mechanism to guarantee bounded access delay and support QoS.

Token rotation time (TRT) and target TTRT

- **TRT (Token Rotation Time):** Actual time a token takes to make a full logical rotation past all stations.
- **TTRT (Target Token Rotation Time):** Configured parameter that sets the maximum acceptable average rotation time, effectively defining how often each station must see the token.

Each station measures TRT and compares it to TTRT: if the token arrives earlier than TTRT, the station can transmit; if later, it must restrict its use to preserve bounded delay.

Token holding time (THT)

Each station computes its **Token Holding Time (THT)** as some function of the difference between TTRT and the observed TRT; this determines how long it may hold the token to send asynchronous data. This dynamic control ensures fairness and prevents a single station from monopolizing the ring.

Traffic classes: synchronous and asynchronous

FDDI supports two traffic classes on the same medium.

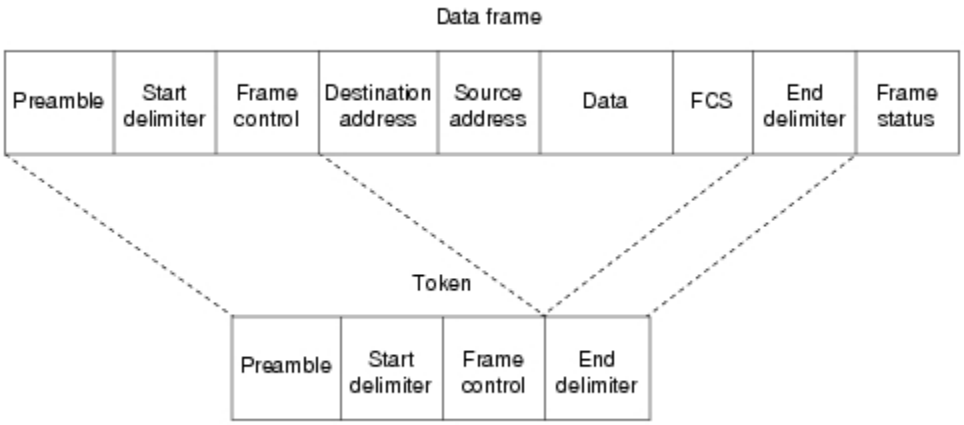
- **Synchronous traffic:**
  - Reserved bandwidth at each token rotation.
  - Used for time-critical flows (voice, video, industrial control).
  - Stations are assigned a synchronous allocation (Si), which they can always send when they get the token, as long as global TTRT is respected.
- **Asynchronous traffic:**
  - Best-effort traffic using remaining capacity after synchronous transmissions.
  - Governed by THT; stations use any leftover time until the token must be released to maintain TTRT.

This **timed token** design gives FDDI better delay control than classic Ethernet's contention-based CSMA/CD.

FDDI frame and token formats

FDDI frame formats are closely related to IEEE 802.5 Token Ring, but adapted for fiber, higher speed, and dual-ring operation.

General data frame format



A typical FDDI data frame contains the following fields:

- **Preamble:**
  - Repetitive pattern for synchronization at the physical level.
- **Start Delimiter (SD):**
  - Marks the beginning of the frame at the MAC level.
- **Frame Control (FC):**
  - Indicates type of frame (e.g., token vs data, synchronous vs asynchronous) and control information.
- **Destination Address (DA):**
  - 2 or 6 bytes; identifies the intended receiver; can be unicast, multicast, or broadcast.

- **Source Address (SA):**
  - 2 or 6 bytes; identifies the sending station.
- **Data:**
  - Payload up to roughly 4500 bytes (maximum around 4352–4550 bytes depending on interpretation), allowing efficient carriage of higher-layer PDUs like IP packets.
- **Frame Check Sequence (FCS):**
  - 4-byte CRC providing error detection.
- **End Delimiter (ED):**
  - Half-byte or one byte code indicating end of data frame.
- **Frame Status (FS):**
  - Contains bits such as A (address recognized), C (frame copied), and E (error).
  - As the frame traverses the ring, receiving stations set these bits to signal whether the frame was correctly received and copied.

After one full ring circulation, the source station removes its own frame from the ring (absorptive removal), preventing continuous circulation.

Token frame format

The token is a special, short frame with no data payload.

- Fields: Preamble, SD, FC (with bit pattern indicating token), ED.
- No addresses or data; simply grants permission to transmit.
- When a station with pending data receives a free token, it absorbs the token, transmits its allowed frames, then releases a new token back onto the ring.